# Ritik Varaganti

San Ramon, California • 227159@students.srvusd.net• varagantiritik@gmail.com

## Education

**Dougherty Valley High School** San Ramon, California
3.94/4.00 | 4.46/5.00 (UW/W) GPA
2023 - 2027
Relevant Coursework: AP Computer Science Principles, AP Calculus AB, AP Computer Science A, Honors Physics, Honors Precalculus, Honors Chemistry

**Diablo Valley College**
4.00/4.00 GPA
Relevant Coursework: Calculus 1, Computer Science 1

## Experience

**Research Intern @ University of North Carolina, QIAI Lab** Ongoing
- Compiled and cleaned 200+ GB of Mechanical Engineering textual data from arXiv, Google Scholar, etc. totaling over 50,000 research papers
- Trained Facebook's LlaMA 4 model in collaboration with UNC Mechanical Engineering department, paper incoming

**Research Intern @ University of California at San Diego, Ujima S&P Lab**
- Utilizing Natural Language Processing Techniques to analyze generated AI content on short-form video platforms

## Projects/Publications

**Behavioral and Emotional Analysis for Comprehensive Observation and Diagnosing for Accurate Early Detection and Intervention in Autism Spectrum Disorder (BEACOD)** 2025
- Research paper recognized as one of the best in state, hand-selected to present at San Francisco State University for the Northern California Junior Science and Health Symposium from the US Department of Defense
- Designed and built a multimodal Machine Learning model that is able to effectively diagnose ASD from an entirely behavioral approach (combines data from facial micro-expressions, audio cues, verbal responses)
- Used techniques like Optical Flow (Lucas-Kanade method), K-means, etc, developed custom-built Transformer and CNN based models on audio, video, and speech data from ASD individuals

**ISIC Skin Cancer Detection with 3D-TBP** 2024
- Devised ensemble model of XGBoost, CatBoost, LGBM run on text based data + Convolutional Neural Network run on image based data
- Used techniques like ensembling models, One Hot Encoding, advanced data analysis (multicollinearity, feature engineering, variable correlations, etc.), voting classifiers, and more
- Achieved scores within top 2% of 13,000+ participants, achieving partial area under the ROC curve (pAUC) score of 0.1853

## Awards

**YSIC Youth International Investing Competition** June, 2024
- Scored in top 10 in 600+ participating teams as graded by 4 Wall Street executives
- Scraped and compiled stock data, ran algorithms (LSTM, BiLSTM, etc.) on metrics like Stochastic Oscillator, RSI to determine optimal stocks, analyzed stock data w/ time series decomposition

**RSNA 2024 Lumbar Spine Classification** October, 2024
- Scored within top 6% in 15,000+ participants
- Developed convolutional neural network to classify condition based on spine photographs

**American Math Competition (AMC)** November, 2023
- Scored within top 1% of test-takers in the United States in the first stage of the American Mathematics Competition
- Achieved Distinguished Honor Roll status (21/25 questions correct)

**Presidential Volunteer Service Award - Gold** July, 2024
- Partnered with local nonprofit to assemble and distribute 15+ efficient NAS data systems for 75+ hours
- Worked as assistant to Autism Paraeducator at local elementary school, taught and cared for children with Autism Spectrum Disorder (ASD) for 80+ hours

## Additional

**Languages:** Python, Java, HTML/CSS, JavaScript

**Frameworks:** PyTorch, Keras, TensorFlow, SciKit-Learn, Hugging Face Transformers, Torchvision, Torchaudio

**Libraries:** Seaborn, matplotlib, LightGBM, XGBoost, CatBoost, Optuna, pandas, NumPy, OpenCV, SciPY, yt-dlp, Librosa